

Lecture 1

Aug 25, 2026

Statistics as a way of Thinking

Instructor: Alexander Volfovsky

Old Chem 223

OH: Wednesday 12^{pm} - 2^{pm}

Kat Husar

Old Chem 203A

OH: Monday 2^{pm} - 4^{pm}

Class time:
Tue Th 10⁰⁵ - 11²⁰
Old Chem 025

Lab time:
Fri 11⁴⁵ - 1⁰⁰ pm
Lab 2

THE HISTORY OF STATISTICS

*The Measurement of Uncertainty
before 1900*

STEPHEN M. STIGLER



Learning objectives

- Explain why statistics is more than a collection of numerical techniques by connecting measurement, mathematical regularity, and uncertainty to the historical development of statistical reasoning.
- Compare external and internal assessments of numerical reliability using the Trial of the Pyx and Cassini examples, including the assumptions behind different rules for how errors accumulate.
- Distinguish repeated measurements of the same quantity from measurements across different people or objects, and identify when combining observations is defensible.

Statistics : what?

- making sense of measurements in populations
- uncertainty in data
- use data for downstream tasks
- How much can you learn about the real world from data

The road to be travelled over . . . is full of interest of its own. It familiarizes us with the measurement of variability, and with curious laws of chance that apply to a vast diversity of social subjects. This part of the inquiry may be said to run along a road on a high level, that affords wide views in unexpected directions, and from which easy descents may be made to totally different goals to those we have now to reach. I have a great subject to write upon. (Galton, 1889, p. 3)

Statistical Science

Bacon 1600ish

Arbutnot 1690s

Kelvin 1910s

Why this book?

Bag of tricks

means, variance, hypothesis testing, probability

Ordinary least squares vs Regression

OLS: a way of estimating a and b while
(1805) making E close to zero

$$E = ax + by, \quad x, y \text{ observed}$$

minimizing $(ax + by)^2$

Regression: looking for a and b that explains

Galton (1885) $y = ax + b$. How? minimizing $(y - ax - b)^2$

How do we trust a numerical assessment?

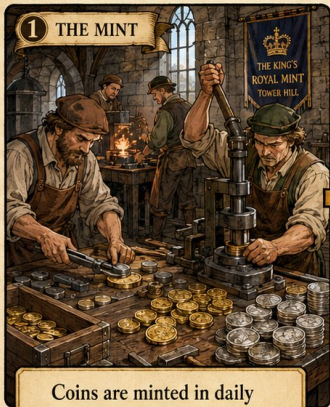
- Repeated experimentation
- Large sample size
- *p-values*
- randomness? when sampling from a population
- quantifying uncertainty
- *appeal to authority*



The Trial of the Pyx

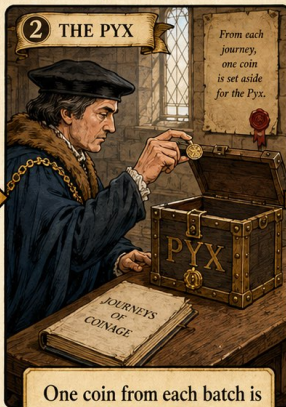


An early sampling inspection system for coinage



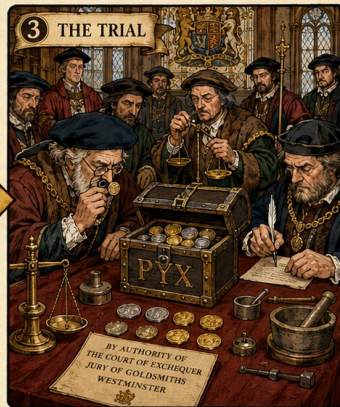
1 THE MINT

Coins are minted in daily batches (called 'journeys').



2 THE PYX

One coin from each batch is set aside in the Pyx box.



3 THE TRIAL

At intervals, a jury opens the Pyx and tests the sample.



Why statisticians care:

This is an early example of sampling inspection and hypothesis testing.



How the sampling worked



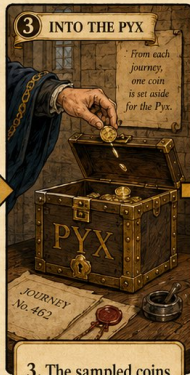
1 THE MINT

1. The Mint produces coins in batches ('journeys').



2 SELECT ONE

2. From each batch, one coin is selected.



3 INTO THE PYX

3. The sampled coins accumulate in the Pyx.



4 PASSAGE OF TIME

4. The box is opened only at intervals — often after 1 to 4 years.



5 THE JURY

5. A jury of goldsmiths inspects the sample.



Key idea:

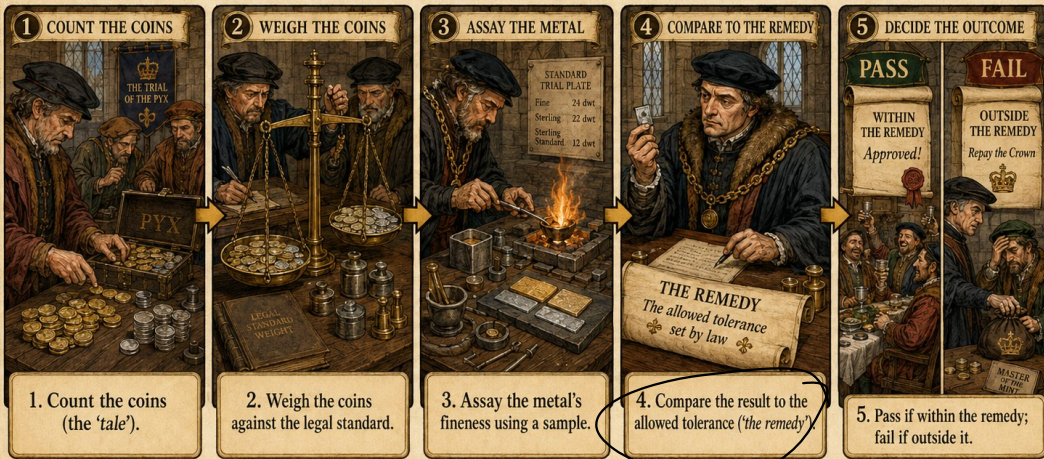
only a sample is examined, not the entire coinage.



The Pyx was a locked strongbox; the sample represented the mint's production over time.



How the trial judged the Mint



Historically, failure could mean repayment to the Crown — or worse.



In modern statistical language, this is a decision rule based on sample evidence.

The lgs measurement/evaluation

How do we judge if our coins are real?

5 grains per coin tolerance

Sampled 100 coins, what is the right tolerance

$$500 \text{ grains} = 5 \times 100$$

$$50 \text{ grains} = 5 \times \sqrt{100}$$

$$250 \text{ grains} = 5 \times 100/2$$

High level outline

- ① The combination of observations in astronomy + geodesy
- ② The use of probability for inferential purposes
- ③ The development + spread of procedures for biology and the social sciences

Combining Observations 101

5 measurements of 1 person's height

averages $\rightarrow$ more accurate measurement/estimate for that person

1 measurement of 5 people's heights

average height in population

Table I.1. Cassini's 1740 comparison of nearly 2000 years' data on the obliquity of the ecliptic with values found by linear interpolation.

Source	Date	From observation
Eratosthenes	230 B.C.	23°51'20"
Hipparchus	140 B.C.	23°51'20"
Ptolemy	140 A.D.	23°51'10"
Pappus	390	23°30'0"
Albategnius	880	23°35'0"
Arzachel	1070	23°34'0"
Prophatius	1300	23°32'0"
Regiomontanus	1460	23°30'0"
Copernicus	1500	23°28'24"
Waltherus	1500	23°29'16"
Danti	1570	23°29'55"
Tycho	1570	23°31'30"
Gassendi	1600	23°31'0"
Cassini	1656	23°29'2"
Richer	1672	23°28'54"
Paris Observatory	1738	23°28'20"

Source: Cassini (1740, p. 113).



Assessment types

External

Internal

today's lecture: 1-8, total covered: 1-8

FOR NEXT CLASS

Read: 9-31

Think about: more data > less data?
When is it ok to combine
observations?